

Making a call. Using this technology, there is no need to physically take out the phone to make a call. Using Sixth Sense, one can call another person in a swift and convenient manner. On using this application, the number pad is displayed on any surface the user wants. When the user positions fingers on displayed numbers, the device automatically understands the numbers that the user is signalling and dials the number (Fig. 4).

Searching the map. This function can be used to plot a course, search for places, locate the most viable paths and provide assistance while navigation, to name a few.

Drawing application. Using this device, one can express artistic skills anywhere, any time. The associate application draws and projects images on available surfaces. Drawings saved in device memory can be retrieved any time.

Taking a picture. The camera is used to take an image of the scene the user is looking at. When four fingertips of the user are combined to form a rectangular frame, the camera understands it as a frame-capturing gesture and performs the appropriate task (Fig. 5).

Clock. Using clock application, one can check the time. When the user puts wrist in front of the camera, a digital watch appears.

Multimedia reading experience. The user can program the device to project related videos onto newspaper articles.

Getting product information. Product information can be obtained by Sixth Sense technology because it uses image-recognition or image-maker technology. As soon as the user picks up a product off the shelf, the system scans the product and feeds



Fig. 4: Making a phone call using Sixth Sense technology

information such as product specification, user ratings, reviews and other relevant information to the user.

Checking emails. The user can check emails by gesturing @ sign. To open the application, the user has to draw @ in the air. The device will understand the gesture and open the application. Mails can be synchronised with the device in a way that the user can view old mails even in offline mode. This application uses the Internet to search, receive and send new, received mails.

Getting flight updates. Once the user scans the boarding pass, the system feeds relevant flight information like its on-time status or any change in the gate entry point.

Advantages and challenges

As the device is small, it is portable. It supports multiple touches and multi-user communication. Tasks can be carried out through the user's motion and gestures. This, to a great extent, also saves the user's time. Data can also be obtained from the Internet by the user.

The device can have a wide variety of applications for different consumers depending on what they imagine and want. This project involves an

open source code, which further helps individuals develop their own applications that are workable on this device. WUW can perform tasks of various other devices by itself and, hence, end up being economical.

Finding an adequate vision-based gesture-recognition algorithm has always been the most demanding part of this device. There are many obstacles involved in the working of the device with precision and usefulness of the gesture-recognition

software. There are also limitations on the equipment used. Programming code is very specific to the particular device, and there may be trouble in executing it on another device.

Methods of further development

Searching for alternatives for colour markers would be one of the ways to further improve this device. Enhancing the device so that it can be utilised in such diverse sectors as gaming, education and entertainment will generate a greater number of consumers. For example, an extra speaker can be added to elevate the experience.

An application that can convert gestures into sound could be developed. And this application, in turn, may be used to help people with certain disabilities communicate more effectively with others.

The problem with vision algorithms for gesture recognition can be rectified by integrating artificial intelligence (AI) with the device. This would help recognise human signals much more efficiently, and errors in misinterpretation of gestures would be greatly reduced.

Conclusion

Sixth Sense technology is the science of the future and has the potential for ushering in a new era in the field of technology. This technology could truly change the way humans interact with the real world and give everyone complete awareness about their environment. Sixth Sense prototype implements several applications that demonstrate its utility, feasibility and practicability. There is no doubt that this technology will continue to grow and may very well become the next big thing. **EFY**



Fig. 5: Frame-capturing gesture for camera (Credit: www.seminarsonly.com)